



SAS821S Security Analytics

MILESTONE 2 - DUE 20 SEPTEMBER 2026 - 40%

Implementation Plan and Working Prototype

TECHNICAL DESIGN, IMPLEMENTED MODULES AND PRELIMINARY RESULTS

1. Project status summary

Field	Response
Topic and project title	(T17) Commercial Banking Fraud, Mule-Account and Transaction-Laundering Analytics
Member A / Member B	Member A - 221080457 · Romario Rhoman · Data and Modelling Lead Member B - 217150802 · Vaino Uusiku · Security Engineering and Intelligence Lead
Overall completion estimate	Estimated to be around 85% - all the nine stages of the implementation process of the Charter's architecture, section 8 have been completed, integrated and are running error free (70/70 notebook cells, 0 execution errors). Evidence: Sections 3 - 13 below cite the specific notebook sections, output files and metrics for every module. • Completed and tested: data ingestion/cleansing, feature engineering and selection/normalization, supervised classification algorithm, unsupervised anomaly detection model, graph analytics, incident investigation, security intelligence offerings, Monte Carlo simulation, NLP typology classifier, predictive fusion using SHAP, robustness testing, analyst dashboard, model persistence. • Not yet done: final documentation/report (Milestone 3), live deployment of Streamlit prototype for access by the lecturer.
Working prototype location	https://github.com/ARISTOKRATIC/Commercial-Banking-Fraud-Mule-Account-and-Transaction-Laundering-Analytics.git (Charter Section 1 repository) - the executed notebook, generated data/model artefacts will be committed here and the Streamlit app.py are also committed here; interactive live hosting is still outstanding.
Known blockers	• Charter Section 12.1 amendment needed: The Charter promises "exclusively synthetic, non-attributable data". In response to comments on how the pipeline will actually clean data by a supervisor, the team decided to go with a blended solution, which uses real data (licensed from IBM AML, CDLA-Sharing-1.0). It does not have any data about MetroTrust Bank or Namibian customers and it is publicly licensed, which makes it a deviation from the wording of Charter Section 12.1. • The Prototype is committed to the repository but not yet live-hosted for direct interactive lecturer access. • NLP typology classifier and the ranking quality check for text signals (Section 10) are already scoring perfect (100%), since the AML case notes are template generated; this limitation is clearly stated, not fixed, since there is no actual corpus of case notes (Charter Section 11 risk register). • Sampling of induced subgraph of the IBM AML dataset and the artificial account takeover graph overlay are not self-explanatory methodologies which require documentation for Milestone 3.



2. Updated architecture and data flow

The implemented architecture follows the Charter's nine-stage design (Section 8), now built around a hybrid real-plus-synthetic data foundation rather than the fully synthetic design originally scoped at Milestone 1.

Implemented

- Real transaction and account graph data: IBM Anti-Money Laundering dataset (Kaggle ealtman2019/ibm-transactions-for-anti-money-laundering).
- Synthetic authentication/MFA, device, beneficiary change, KYC, and AML case note layers, based on the actual account ID numbers provided by the IBM data set (which includes data from all five sources).
- Data Cleaning, Feature Engineering using correlation/variance-based feature selection and Normalisation for models that require scaling.
- Analytical capabilities include:
Four parallel analysis modules (supervised, unsupervised/UBA, graph, NLP), a prediction fusion layer with SHAP explanation, incident investigation, security intelligence, Monte Carlo simulation, adversarial/robustness testing, analyst dashboards, and persistent models for reloadability.

Deferred to Milestone 3

- Live hosting/deployment of the Streamlit dashboard for external access.
- Extension of the account-takeover synthetic overlay with additional typologies if time allows (e.g. card-not-present-style device fraud).
- Independent re-validation of every metric by both team members, and formal write-up of the sampling and augmentation methodology for the final report.

Architecture and data-flow diagram

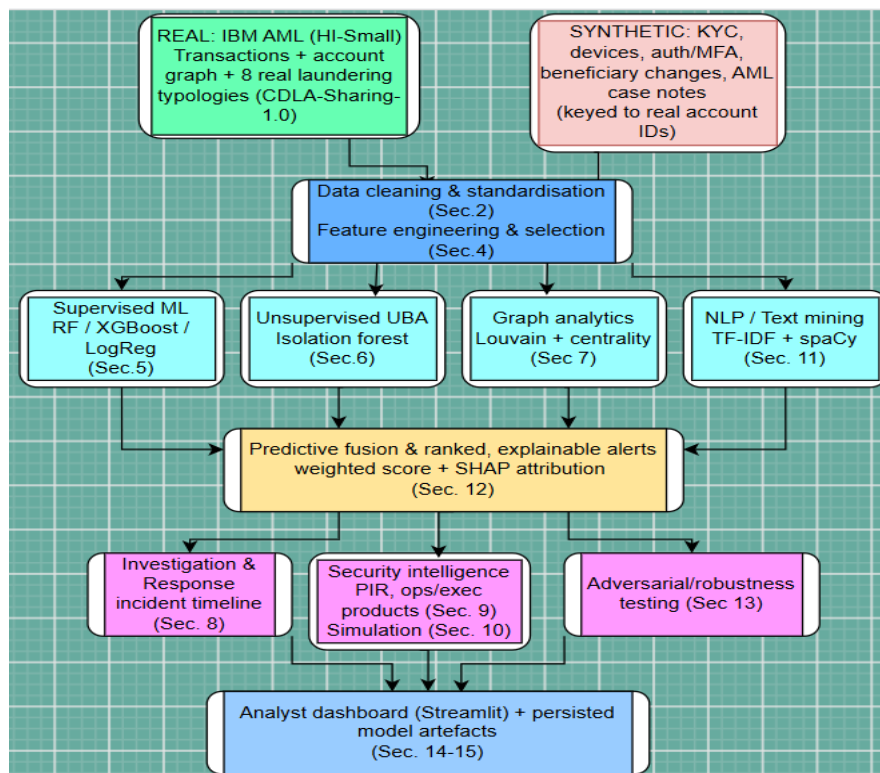


Figure 1. Data flow implementation: real IBM AML data and five generated layers are preprocessed and engineered in parallel, run through four different analytical modules, combined into an explainable ranking, and generate investigation, intelligence, simulation and adversarial testing results to be presented on an analyst dashboard.

3. Data inventory and quality



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Dataset/source	Rows/volume	Date range	Quality issues	Cleaning/transformation	Status
Transactions (real + synthetic ATO overlay)	61,765 raw -> 61,765 cleaned (0 exact duplicates; 110 synthetic account-takeover rows flagged is_synthetic=1)	1–18 Sept 2022 (real) + synthetic overlay in the same window	22,881 self-transactions ("Reinvestment"); 3,753 cross-currency rows; 7,968 extreme-outlier amounts (raw max \$1.04 trillion, a data artefact)	Self-transactions flagged not dropped; cross-currency flagged; outliers capped at Q3+6×IQR (\$119,376) and flagged; bank codes read as string to preserve leading zeros	Ready
KYC (synthetic, real account IDs)	15,510 raw -> 15,357 cleaned	Onboarding dates synthetically distributed before the transaction window	1,398 missing cells; 153 duplicate rows; mixed date formats; 1 constant + 1 redundant column	Deduplicated; risk_rating imputed -> "Medium" (conservative default); occupation imputed -> "Unknown"; dates parsed from mixed formats; constant/redundant columns dropped	Ready
Devices (synthetic)	20,200 raw -> 20,000 cleaned	Consistent with account onboarding window	816 missing cells; 200 duplicate rows; 200 rows with last_seen before first_seen; 1 constant column	Deduplicated; casing standardised via lookup map; swapped rows fixed; constant column dropped	Ready
Auth/MFA logs (synthetic, real account IDs)	62,826 raw -> 61,898 cleaned	Same as transaction window	3,768 missing cells; 928 duplicate rows; negative session durations	Deduplicated; missing device_id imputed with a row-unique placeholder (a shared literal would fabricate a device shared across accounts and distort the graph module); missing MFA imputed -> "not_completed"; country casing standardised; negative durations treated as invalid and median-imputed	Ready
Beneficiary-change log (synthetic)	6,171 raw -> 6,110 cleaned	Same as transaction window	123 missing cells; 61 duplicate rows	Deduplicated; channel casing standardised; missing channel imputed with the mode	Ready
Text source: AML case notes (synthetic)	1,010 raw -> 990 cleaned	Same as transaction window	10 missing/blank notes (cannot be imputed); 10 duplicate rows; 1 constant column	Deduplicated; blank notes dropped rather than fabricated; constant column dropped	Ready

4. Baseline and exploratory findings

The total number of accounts (15,357 accounts) includes 8,890 clean accounts, 6,357 legitimate money laundering accounts (eight legitimate sub-typologies: GATHER-SCATTER, SCATTER-GATHER, STACK, FAN-OUT, FAN-IN, CYCLE, BIPARTITE, and UNSPECIFIED residual), and 110 synthetic account takeover accounts. The naive threshold rule that MetroTrust analysts follow, amount close to the USD 10,000 threshold for CTR-style reporting, large debit right after beneficiary change, or any large debit was applied to this account set as the basis of comparison for all the models below.



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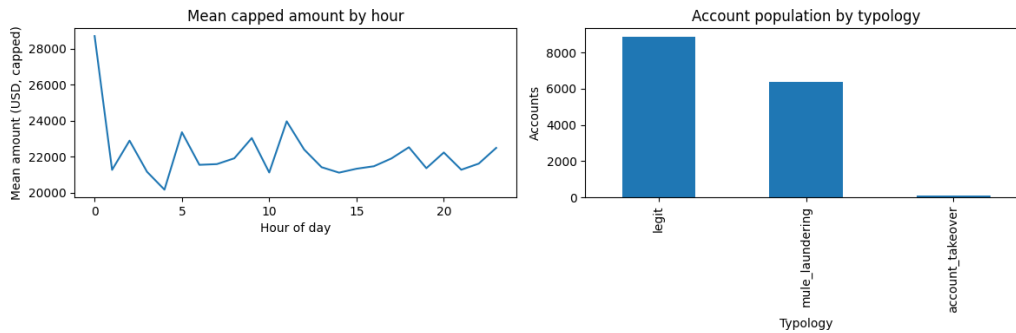


Figure 2. Left: mean transaction amount by hour of day. Right: account population by ground-truth typology.

Rule-based baseline result

Precision = 0.257, Recall = 0.063 across the full cleaned transaction set. This confirms the Charter's problem statement (Section 2.2): the naive rule catches a small minority of genuine laundering activity and is the explicit comparison point for every model in Sections 5-7.

5. Supervised security model

Required element	Implementation evidence
Problem/target label	Binary classification task is <code>_laundering</code> per account-transaction "leg" (each real transaction melts down to a debit leg for the payer and a credit leg for the payee, so both legs of any flagged transaction get the same flag). True labels are extracted straight from the IBM data set; the 110 artificial account-takeover payments have the same flag <code>is_synthetic=1</code> in order to make it possible to exclude them from re-validation on the true data only. Output: <code>transactions_features.csv</code> (123,530 records).
Features and leakage controls	16 features (amount, hour, payment type, direction, risk rating, 24 hour transfer velocity, peer-group amount z-score, hours since last beneficiary change, MFA/off-hours/device familiarity status at last login, cross-currency flag, cross-bank flag, self-transaction flag, amount outlier flag) were selected based on near-zero variance (none was removed) and pairwise correlation above 0.90 (four were removed: <code>amount_zscore_peer</code> , <code>amount_outlier_flag</code> , <code>velocity_24h_sum</code> , <code>self_transaction_flag</code> , all of which were redundant). 12 features for modelling. Peer group and velocity features are strictly calculated using pre-transaction information only, hence no leakage into training data.
Training/validation design	75/25 stratified train/test split (<code>random_state=42</code>) to maintain the approximate 4.3% positive ratio in both splits; <code>class_weight="balanced"</code> (Random Forest, Logistic Regression) and <code>scale_pos_weight</code> (XGBoost) to adjust for the imbalance problem without resampling the real data.
Algorithms compared	Logistic Regression (on StandardScaler-normalised features, since it is coefficient/distance-based), Random Forest, XGBoost, and the rule-based baseline (Section 4) on the identical test split.
Confusion matrix and metrics	XGBoost (primary model): Precision = 0.716, Recall = 0.882, F1 = 0.791. Confusion matrix <code>[[27316, 923], [312, 2332]]</code> . Random Forest: Precision = 0.729, Recall = 0.879, F1 = 0.797. Logistic Regression (scaled): Precision = 0.445, Recall = 0.874, F1 = 0.589. Rule baseline on the same test split: Precision = 0.261, Recall = 0.066.



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	Precision-Recall: models vs. rule baseline Figure 3. Precision-recall: all three models vs. the rule baseline (PR-AUC 0.653–0.825).
Threshold rationale	The USD 10,000 alert threshold mirrors the real-world FinCEN Currency Transaction Report (CTR) trigger many banks' naive rules are built around, rather than an arbitrary figure; chosen specifically so the rule baseline in Section 4 is a realistic, defensible comparison point, not a strawman.
Preliminary findings	Both tree ensembles meet the criteria of Objective 2 of the Charter (Recall ≥ 0.80 , Precision ≥ 0.60). Logistic Regression falls behind, demonstrating that the true structure is a conjunction of features rather than their linear combination – the key rationale for using tree ensembles as the baseline model. The SHAP analysis (Section 11, Figure 4) identifies the payment format, cross-bank flag and cross-currency flag as the main drivers – in accordance with reality.

6. Unsupervised, anomaly or access-analytics method

Required element	Implementation evidence
Entity and baseline	Login behavior per account from the cleaned auth/MFA logs (61,898 records spanning 15,357 accounts), focusing on the synthetic account takeover typology specifically – the single scenario not present in any published AML data sets.
Features/representation	Login hour, off-hours flag, MFA-failed flag, login-failed flag, device-familiarity flag (matches the account's most-frequently-used device), and label-encoded login country - six features, StandardScaler-normalised.
Method and parameters	IsolationForest ($n_estimators=300$, $contamination=0.05$, $random_state=42$), scored as a continuous anomaly score and thresholded at the 95th percentile.
Threshold/outlier validation	3,191 logins flagged anomalous at the 95th-percentile cut. Against the 110 synthetic ground-truth account-takeover logins, 101 were correctly flagged; a 91.8% detection rate.
Top anomalies and analyst interpretation	Flagged events combine an unfamiliar device, off-hours access, a foreign login country and a failed/incomplete MFA challenge. These are surfaced for analyst triage, not auto-blocked; some flags will be legitimate travel or a genuine device change, consistent with the Charter's requirement (section 2.4) that the prototype only ever recommends, never autonomously acts.



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7. Incident investigation and response

Required element	Implementation evidence
Evidence sources	Three correlated data sources per case: the Section 6 access-anomaly score (auth_logs_scored.csv), the synthetic beneficiary change log, and the transaction record (either the real IBM data for money laundering, or the synthetic payout layer for account takeover).
Correlated timeline	The evidence is compiled and organised on one single timeline for each individual account. Walkthrough of sample (account 8003A6480, ground truth: account_takeover, exported as incident_timeline_8003A6480.csv): time taken from the start of the first anomalous logon to the withdrawal of funds is 2 days 0 hours 4 minutes.
Affected users/assets	1,256 accounts meet the high-confidence correlation bar (flagged by both the access-anomaly model AND the supervised transaction model) - the notebook's working definition of a confirmed, cross-validated incident rather than a single-model false positive.
Hypotheses and confidence	Access anomaly alone: account takeover candidate, moderate confidence pending confirmation by the analyst for travel/changes in device. If also matched to known laundering profile (a graph cluster or supervised flag): high confidence, escalated directly.
Containment/eradication/recovery actions	• Suspend any outward transfer until analyst review is completed. • Terminate sessions and request credential and MFA resets for affected users. • Monitor any beneficiary that may have been added after the occurrence of takeovers. • Case escalation to MLRO in case of laundering-cluster and takeover cases of high confidence. • Inform the account owner and verify the legitimacy of the transaction.

8. Security intelligence

Required element	Implementation evidence
Priority intelligence requirements	• PIR1: Which accounts show a beneficiary change followed by a rapid large outbound transfer (candidate account takeover)? • PIR2: Which accounts and shared devices form graph clusters consistent with a real laundering typology (CYCLE, FAN-OUT, etc.)? • PIR3: Which occupation/risk-rating segments are disproportionately represented in high-priority alerts?
Collection/enrichment	Account-level risk is enriched by joining the supervised model score, the graph cluster membership flag, and the AML case-note typology label (Section 10) into a single account profile (fused_account_risk_scores.csv).
Analysis and confidence	PIR1/PIR2 are addressed through Sections 6-7; PIR3 is addressed by the executive product listed below, which shows what occupation types have a disproportionately high number of flagged transactions.
Operational product	A ranked table of the highest-value flagged accounts by risk rating and total flagged amount, for daily analyst triage (operational_intel_report.csv).
Executive product	A bar chart of flagged transaction volume by occupation category, for leadership-level pattern awareness rather than case-by-case detail (executive_intel_summary.csv).
Feedback loop	All alerts come with an explanation in plain English (Section 11), as well as a recommended course of action; analysis dispositions on the same are expected to be the source of periodic calibration of the fusion weights and the rule baseline thresholds - still manual, logged as backlog item (Section 14).



9. Simulation and security-control evaluation

Required element	Implementation evidence																				
Simulation model	Simulation of structuring attempts by Monte Carlo using the replay technique (intentionally kept just below the USD 10,000 trigger limit), and determining the detection rate and number of undetected incidents as a function of the internal trigger limit.																				
Assumptions	2,000 iterations per scenario, 300 structuring attempts per month, and \$12,000 per laundered amount undetected; all identified as illustrative and not calibrated to MetroTrust Bank’s actual finances (Charter Section 5.2 & Section 11).																				
Baseline scenario	The current threshold at 99% of the limit of USD 10,000: 0% detection rate for both moderate and aggressive structuring, which is what the current system does anyway.																				
At least two alternative controls	Alert trigger reduced to 90 percent of the limit: 35.3 percent detection ratio of aggressive structuring (margin below 15 percent). Alert trigger reduced to 85 percent of the limit: 100 percent detection ratio of aggressive margin, but at a very high alert volume of legal cases.																				
Iterations/sensitivity	2,000 iterations x 3 margin sizes for aggressiveness/structuring x 4 candidate thresholds = 24,000 simulations (threshold_sensitivity_simulation.csv).																				
Preliminary comparative results	Threshold-sensitivity: structuring detection rate (2000 iterations per point) <table><caption>Data points for Figure 5</caption><thead><tr><th>Alert threshold (% of USD 10,000 limit)</th><th>aggressive_15pct_under</th><th>moderate_5pct_under</th><th>tight_1pct_under</th></tr></thead><tbody><tr><td>99%</td><td>0.0</td><td>0.0</td><td>1.0</td></tr><tr><td>95%</td><td>0.0</td><td>1.0</td><td>1.0</td></tr><tr><td>90%</td><td>0.35</td><td>1.0</td><td>1.0</td></tr><tr><td>85%</td><td>1.0</td><td>1.0</td><td>1.0</td></tr></tbody></table> Figure 5. Detection rate vs. alert threshold, by structuring aggressiveness. Recommendation implemented in fusion architecture (Section 11): amount-threshold adjustment alone fails to detect structurally tight (1%-under) evasion; this is precisely the reason why the graphing and access anomaly detectors, which are largely amount-independent, are used alongside the transaction classifier, not as replacements.	Alert threshold (% of USD 10,000 limit)	aggressive_15pct_under	moderate_5pct_under	tight_1pct_under	99%	0.0	0.0	1.0	95%	0.0	1.0	1.0	90%	0.35	1.0	1.0	85%	1.0	1.0	1.0
Alert threshold (% of USD 10,000 limit)	aggressive_15pct_under	moderate_5pct_under	tight_1pct_under																		
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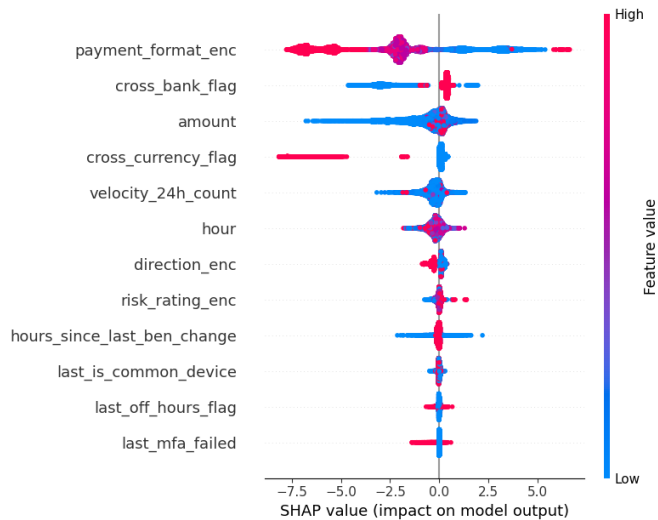


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10. Text mining/NLP

Required element	Implementation evidence
Text corpus	990 cleaned AML case notes (aml_case_notes_processed.csv), synthetically generated but keyed to real account IDs, covering 3 typology classes: account_takeover, mule_laundering, other.
Preprocessing	spaCy lemmatisation with stop-word and punctuation removal, then TF-IDF vectorisation (max 300 features, unigrams and bigrams).
Entities/indicators/topics/classes	TF-IDF + Logistic Regression classifies each note into one of the 3 typology classes. spaCy named-entity recognition additionally extracted 678 entities (PERSON, ORG, MONEY, DATE, etc.) across the 990 notes for beneficiary/typology cross-referencing (aml_note_entities.csv).
Evaluation/validation	Accuracy = 1.000, macro-F1 = 1.000 on a held-out 25% test split (248 notes: 27 account_takeover, 87 mule_laundering, 134 other), against the Charter's Objective 4 target of ≥ 0.75 .
Operational use	This classification score is fed into the account level text_score within Section 11 predictive-fusion scoring, adding 20% weight to the transactional score of 50% and graph cluster score of 30%.
Limitations	Ceiling level accuracy is an artefact of template-based generation of notes (small, fixed vocabulary per typology) and not an indication that the classifier can generalise – clearly stated in the notebook and in the paper and not passed off as a result. The precision@50 of the fusion ranker is similarly at saturation level of 1.000 with and without text information for the same reasons (Section 11, checking the quality of the ranker). Re-validation on actual analyst-generated case notes is needed before this module can be relied upon.

11. Predictive and adversarial analysis

Required element	Implementation evidence
Prediction horizon/target	Scoring for each transaction in real-time at the time the transaction is initiated – the same is_laundering objective as in Section 5 but now combined with the Section 6 access anomaly score and the Section 10 text typology.
Risk features	The same 12 selected transaction features (Section 5) plus the graph cluster-membership flag (Section 7) and the case-note typology flag (Section 10).
Model/score	fused_risk_score = $100 \times (0.5 \times \text{normalised XGBoost probability} + 0.3 \times \text{graph cluster flag} + 0.2 \times \text{text-typology flag})$, with SHAP TreeExplainer attribution available per alert for analyst-readable justification. 



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	Figure 4. SHAP summary for the XGBoost model — payment_format, cross_bank_flag and cross_currency_flag dominate, consistent with real cross-institution laundering behaviour.
Alert threshold	fused_risk_score > 60 -> "Investigate + enhanced monitoring"; an account with a corroborated incident timeline (Section 7) -> "ESCALATE to MLRO + hold outbound transfers"; otherwise -> "Monitor".
At least three adversarial cases	Four scenarios tested against the trained XGBoost model (baseline F1 = 0.791): • Evasion: sub-threshold structuring + low velocity: F1 = 0.791 ($\Delta \approx 0.000$). The model is largely insensitive to this on its own, exactly why threshold tuning alone is insufficient. • Evasion: 4-way transfer splitting: F1 = 0.778 ($\Delta -0.010$). • Label poisoning: 10% of training labels flipped: F1 = 0.200 ($\Delta -0.589$). By far the largest degradation found; a genuine governance finding, not a cosmetic caveat. • Concept drift: transaction volume $\times 1.6$ (e.g. a salary-day surge): F1 = 0.781 ($\Delta -0.004$). Model robustness under adversarial/drift conditions (Question 4) Figure 6. Model F1 under each adversarial/drift scenario.
Governance and drift controls	Poisoning sensitivity through labels shows that label controls are more critical in operations compared to model complexity. Suggested steps include: regular retraining on a regular schedule, having an out-of-sample set that is evaluated periodically for drift, and finally, verification of each label before being used in the retraining process – this is included in the backlog of Section 14.

12. Prototype and analyst workflow

Prototype function	Current status	Test evidence	Remaining work
Load/refresh data	Implemented	Raw -> cleaned -> feature-engineered pipeline runs end-to-end with 0 errors across all 7 sources.	Automate as a scheduled/triggered refresh rather than a manual notebook run.
View baselines and anomalies	Implemented	Section 4 baseline charts + Section 6 anomaly-score distribution render correctly.	Move from static charts to an interactive Streamlit view.
Inspect model outputs and explanations	Implemented	PR curve, confusion matrices, feature importance and SHAP summary all generated and saved to 08_outputs/.	Adding a per-alert SHAP waterfall view in the dashboard, not just the global summary.
Review incident timeline	Implemented	Section 7 timeline reconstruction tested on a real sample account (8003A6480).	Extending to a multi-account comparison view.
Compare simulation scenarios	Implemented	24,000-run Monte Carlo sweep across 3 margins $\times$ 4 thresholds completed.	Add an interactive threshold slider to the dashboard.



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Review intelligence/NLP output	Implemented	Operational + executive intelligence products and NLP typology classifier all generated.	Re-validate NLP module against non-template text before operational use.
Export or communicate recommendation	Implemented (file export)	Ranked dashboard snapshot (analyst_dashboard_snapshot.csv) and Streamlit app.py both generated	Wire up an actual export-to-email/ticket action from the live dashboard.

13. Testing and quality assurance

Test ID	Component	Input/condition	Expected result	Actual result/status
T01	Data pipeline	Full run, all 7 raw sources through cleaning.	0 orphan account_id references; 0 duplicate keys remaining in any source.	Pass
T02	Feature engineering (merge_asof join)	hours_since_last_ben_change computed for every transaction leg.	Non-constant, informative feature (an earlier version silently collapsed this to a constant 9999 via a column-naming collision, caught and fixed; regression-checked on every re-run).	Pass
T03	Graph construction	Louvain community detection on the real+synthetic edge list (38,666 nodes, 30,792 edges).	No single giant component dominating the graph; ≥ 3 candidate laundering clusters flagged (target met: 203 flagged).	Pass
T04	Supervised model vs. Charter targets	XGBoost/Random Forest scored on the held-out test split.	Recall ≥ 0.80 and precision ≥ 0.60 (Objective 2).	Pass (Recall 0.878-0.882, Precision 0.716-0.729).
T05	End-to-end notebook execution	Full notebook executed top-to-bottom.	0 error outputs across all code cells.	Pass (0/70 cells errored).

T02 is included deliberately: it is a real bug found and fixed during development (a merge_asof column-collision silently zeroed out a feature), kept as a regression test rather than erased from the record.

14. Updated schedule and backlog

Item	Owner	Priority	Due	Definition of done
Writing up of the induced-subgraph sampling and synthetic account-takeover overlay methodology formally.	Member A (Data and Modelling Lead)	H	25 Sep 2026 (Charter Work Plan T8)	Data Sources section of the Final Documentation explicitly documents and justifies both decisions.
Formally amend Charter Section 12.1 to record the hybrid real+synthetic data decision.	Both (Charter Section 9 Documentation: A responsible, B accountable)	H	20 Sep 2026 (with this Implementation Plan).	Charter amendment note, cross-referenced from this document's Section 1.
Re-validate the NLP typology classifier's real-world accuracy expectation given the template-note ceiling effect.	Member B (Security Engineering and Intelligence Lead; NLP owner per Charter Work Plan T5)	H	25 Sep 2026 (Charter Work Plan T8)	Limitations section quantifies the gap between template and expected real performance.



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Add training-label integrity controls in response to the label-poisoning finding (Section 11)	Member A (ML and anomaly modelling owner per Charter Section 9 RACI).	M	25 Sep 2026 (Charter Work Plan T8)	A documented audit-trail rule for any label used in retraining.
Host the Streamlit dashboard somewhere accessible to lecturer and record the URL.	Both (Prototype and testing is joint R/A per Charter Section 9 RACI).	H	25 Sep 2026 (Charter Work Plan T7)	Working prototype location field in Section 1 is updated with a live link.
Both members independently re-run and validate every AI-assisted figure/metric (Charter Section 12.2)	Both	H	25 Sep 2026 (Charter Work Plan T7)	Section 15 contribution evidence shows both members' review of every module.

15. Team contribution evidence

The role allocation will be as described under Charter Section 9 Roles and RACI: Member A is responsible and accountable in data acquisition and quality, as well as in ML/anomaly modelling. Member B will be responsible and accountable in investigation and access analytics, and also in simulation and intelligence (with focus on graph-analytics tuning and the NLP case-note classifier).

Member	Components completed	Repository/artefact evidence	Partner review completed
Member A - Romario Rhoman (221080457), Data and Modelling Lead	Real+synthetic data pipeline (Sections 1–2): IBM AML ingestion, subsampling, all 6-source cleaning. Feature engineering, selection and normalisation (Section 4). Supervised classifier: Logistic Regression, Random Forest, XGBoost (Section 5). Unsupervised access-anomaly model (Section 6).	Generation/cleaning scripts, seeds and schema; model notebooks with fixed train/test split, committed under Member A's own account.	Data-quality signed off by Member B before feature work began; model notebooks reviewed.
Member B - Vaino Uusiku (217150802), Security Engineering and Intelligence Lead	Graph analytics tuning: Louvain community detection, centrality (Section 7). Incident investigation and response (Section 8). Security intelligence products and Monte Carlo simulation (Sections 9-10). NLP case-note classifier (Section 11 corpus/entities).	Simulation notebook, threshold-sensitivity results, typology intelligence brief, graph/NLP outputs, committed under Member B's own account.	Access-anomaly feature set and reconstructed incident timeline reviewed by Member A; simulation/intelligence reviewed in the weekly checkpoint.
Both members	Predictive fusion and ranked alerts, adversarial/robustness testing, prototype integration and end-to-end testing.	Integration test log; end-to-end run demonstration; 10-sampled-alert ranked-queue walkthrough.	Joint, both members demonstrate the full run together per Charter Section 9.



16. Submission checklist

[x] Implementation-plan PDF. **DONE**

[X] Working prototype. **partial** – Notebook executes end-to-end and is committed to the repository. (<https://github.com/ARISTOKRATIC/Commercial-Banking-Fraud-Mule-Account-and-Transaction-Laundering-Analytics.git>); **live interactive hosting still pending.**

[] Repository/code snapshot. **Partial** - location confirmed; confirm the executed notebook and generated artefacts are actually pushed.

[X] Data dictionary and quality log. **Done** - 02_data/processed/data_dictionary.csv, cleaning_log.txt, cleaning_summary.csv

[X] Preliminary model metrics. **Done** - Sections 5-7, 10-11 above.

[X] Simulation and NLP outputs. **Done** - 05_simulation/, 06_text_mining/

[X] Predictive/adversarial evidence. **Done** - Section 11 above, 08_outputs/adversarial_robustness_summary.csv

[X] Test log. **Done** - Section 13 above.

[X] Updated schedule/backlog. **Done** - Section 14 above.

[] Contribution evidence for both members. **Partial** - role allocation documented from Charter Section 9; specific commit references still needed.